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DIAPER WITH COLLECTION POUCH

Abstract

A diaper comprises an outer side and an opposite inner side comprising an absorbent part. A top layer is sealed to the inner surface about its peripheral border and defines a top layer area delimited by the peripheral border that overlies at least a portion of the absorbent part but is not unconnected thereto. The top layer area, the absorbent part and the peripheral border define a pouch therebetween. The pouch defines a pouch cavity therein. The top layer area defines a body contacting surface and provides an opening for accessing the pouch cavity. When the diaper is worn by a user, the opening is provided to be positioned about the anus of the user for providing for faecal matter to enter the pouch cavity during defecation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority on U.S. Provisional Patent Application Ser. No. 63/337,491 filed on May 2, 2022 and incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure generally relates to diapers. More particularly but not exclusively the present disclosure relates to a diaper with a collection pouch.

BACKGROUND

[0003] Diapers include both disposable baby diapers and adult diapers. Diapers can be necessary for adults with various conditions, such as incontinence, mobility impairment, severe diarrhea or dementia. A diaper is a type of underwear that allows the wearer to defecate or urinate without the use of a toilet, by absorbing or containing waste products to prevent soiling of outer clothing or the external environment. When diapers become soiled, they require changing, generally by a second person such as a parent or a caregiver. Diapers have a facing layer to be brought into contact with the wearer's skin, an absorbent panel adjacent thereto, and a water-impervious or a water repellant outer layer. Following defecation, prolonged contact of feces with the wearer's skin may have adverse health consequences such as rashes or infections. For caregivers, changing the soiled diapers of adult patients poses many practical challenges as the task is usually extremely messy with faecal matter spilling out of the diapers and onto the sheets of the patient's bed. The ordeal worsens with patients suffering with severe diarrhea and other digestive issues.

[0004] The present application is concerned with improvements to the diaper collection pouch as described in International Patent Application No. PCT/CA2020/000028 filed on Mar. 16, 2020, and incorporated herein by reference in its entirety. The foregoing document teaches a diaper that comprises a bottom layer and a top layer mounted thereto. The bottom layer defines a top surface and an opposite outer surface and comprises an absorbent core therebetween. The top layer is mounted to the bottom layer about a peripheral border thereof. The top layer defines a top layer area delimited by the peripheral border overlying the top surface and being unconnected to the bottom layer. The top layer area, the top surface and the peripheral border define a pouch therebetween. The top layer area defines a body contacting surface and an opening for providing access to the pouch. When the diaper is worn by the user, the opening is positioned about the anus of a user for providing for faecal matter to enter the pouch during defecation.

OBJECTS

[0005] An object of the present disclosure is to provide a diaper with a collection pouch.

SUMMARY

[0006] In accordance with an aspect of the present disclosure, there is provided a diaper comprising: an outer side and an opposite inner side comprising an absorbent part; and a top layer defining a peripheral border thereof and being sealed to the inner side about the peripheral border thereof and defining a top layer area delimited by the peripheral border overlying at least a portion of the absorbent part and being unconnected thereto, the top layer area, the absorbent part and the peripheral border defining a pouch therebetween, the pouch defining a pouch cavity therein, the top layer area defining a body contacting surface and providing an opening for access to the pouch cavity; wherein when the diaper is worn by a user, the opening is provided to be positioned about

the anus of the user for providing for faecal matter to enter the pouch cavity during defecation

[0007] In an embodiment, the opening is provided along a central line area of the top layer area.

[0008] In an embodiment, the top layer area comprises adhesive at each lateral side of the opening, the adhesive providing for adhering the user's skin at an inner side of each buttock within the intergluteal cleft thereby bringing the opening in close proximity to the anus to receive faecal matter directly therein. In an embodiment, the adhesive is in the form of a pair of longitudinal strips along a longitudinal length of the top layer area so as to be removably sealed along a longitudinal length of the user's skin. In an embodiment, the adhesive provides a seal with the user's skin about each lateral side of the anus and the opening substantially blocking faecal matter from flowing onto an outer surface of the top layer area beyond the seal thereby allowing faecal matter to flow into the opening. In an embodiment, the adhesive for adhering the user's skin at an inner side of each buttock within the intergluteal cleft is covered by a removable liner prior to use.

[0009] In an embodiment, the diaper comprises adhesive on the top layer for providing the top layer for adhering to the user's skin directly on each buttock outside the intergluteal cleft so as to maintain the diaper in position during movement of the user. In an embodiment, the adhesive for adhering to the user's skin directly on each buttock outside the intergluteal cleft is provided in the form of elongate strips on the top layer for providing substantial areas of the top layer without any adhesive. In an embodiment, the adhesive for adhering to the user's skin directly on each buttock outside the intergluteal cleft is covered by a removable liner prior to use.

[0010] In an embodiment, the top layer area comprises a tear line for being selectively torn along a length thereof to selectively form the opening. In an embodiment, the tear line is formed along a central line area of the top layer area for being selectively torn along a desired position of the central line area and along a desired position of the central line area. In an embodiment, the tear line comprises weak lines or perforations, the tear line defining a central portion interposed by front and rear portions, wherein the central portion has a greater number of the weak lines or the perforations than either one of the front and rear portions.

[0011] In an embodiment, the top layer provides for allowing a front part of the absorbent part to be uncovered and thus in direct contact with the user for receiving urine discharge to be absorbed, thereby substantially avoiding urine discharge from into the pouch and mixing with the feces collected therein. In an embodiment, the pouch defines a front part and a rear part, the front part defining a pair of front separate fingers defining an inwardly recessed portion therebetween exposing the front portion of the absorbent part in direct contact with the user for receiving urine discharge to be absorbed.

[0012] In an embodiment, the top layer comprises adhesive for adhering to the users skin and for adhesively sealing the inner side when folding the diaper after use thereby sealing the pouch containing the collected faecal matter.

[0013] In an embodiment, the inner side comprises a longitudinal tension along the length thereof that is greater than the a lateral tension along the width thereof.

[0014] In an embodiment, the top layer area defines a length thereof and forms waves defining longitudinal peaks and valleys along the length thereof, wherein the opening is provided at a selected one of the longitudinal peaks.

[0015] In an embodiment, the peripheral boundary of the top layer defines lateral sides thereof, wherein at least a portion of the lateral sides of the peripheral boundary are cut in a zigzag configuration.

[0016] In an embodiment, the top layer comprises soft elastic elements running along a length thereof.

[0017] In an embodiment, the top layer area comprises silicone circumscribing the opening.

[0018] Other objects, advantages and features of the present disclosure will become more apparent upon reading of the following non-restrictive description of illustrative embodiments thereof, given by way of example only with reference to the accompanying drawings.

[0019] BRIEF DESCRIPTION OF THE DRAWINGS

[0020] In the appended drawings:

[0021] FIG. 1 is a top, perspective view of the diaper in an open position showing the inner side thereof in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0022] FIG. 2 is a top and side, perspective view of the diaper as shown in FIG. 1 in a partially exploded position thereof showing the top layer sheet being separate from the inner side thereof in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0023] FIG. 3 is a side perspective view of the diaper of FIG. 1 in a closed position in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0024] FIG. 4 is a transparent view of the diaper as shown in FIG. 3 following use to show a perspective view of the pouch when filled in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0025] FIG. 5 is a top, perspective view of the diaper in an open position showing the inner side thereof including tear lines and adhesive in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0026] FIG. 6 is an enlarged view of FIG. 5;

[0027] FIG. 7 is a schematic view of the central line area of the pouch of FIGS. 5 and 6 including the tear line in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0028] FIG. 8 is a schematic view of the pouch of the diaper positioned about the buttocks of a user in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0029] FIG. 9 is a top, perspective view of the diaper of FIG. 5 including removable liner strips on the adhesive on the inner side thereof in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0030] FIG. 10 is a top, perspective view of the diaper of FIG. 5 including removable liner sheets on the adhesive on the inner side thereof in accordance with a non-restrictive illustrative embodiment of the present disclosure

[0031] FIG. 11 is a schematic top view representation of the diaper in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0032] FIG. 12 is a schematic cross-sectional representation of the top layer of the diaper of FIG. 11 under stretching tension and showing the portion being cut for creating the opening leading to the pouch of the diaper;

[0033] FIG. 13 is a top plan view of the diaper in in accordance with a non-restrictive illustrative embodiment of the present disclosure;

[0034] FIG. 14 is a schematic top view representation of the diaper in accordance with a non-restrictive illustrative embodiment of the present disclosure; and

[0035] FIG. 15 is a schematic top view representation of the diaper in accordance with a non-restrictive illustrative embodiment of the present disclosure.

Description

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0036] Generally stated and in accordance with an embodiment, there is provided a diaper comprising an outer side and an opposite inner side comprising an absorbent part. A top layer is sealed to the inner surface about its peripheral border and defines a top layer area delimited by the peripheral border that overlies at least a portion of the absorbent part but is not unconnected thereto. The top layer area, the absorbent part and the peripheral border define a pouch therebetween. The pouch defines a pouch cavity therein. The top layer area defines a body contacting surface and provides an opening for accessing the pouch cavity. When the diaper is worn by a user, the opening is provided to be positioned about the anus of the user for providing for

faecal matter to enter the pouch cavity during defecation.

[0037] With reference to the Figures, non-restrictive illustrative embodiment of the present disclosure will now be described.

[0038] With reference to FIGS. **1**, **2** and **3**, the diaper **10** includes an inner side **11** and an outer side **13** thereof. FIG. **1** show the diaper in the open position and FIG. **3** shows the diaper in the closed position when fitted on the user.

[0039] The diaper **10** has front and rear ends, **12A** and **12B**, respectively, and opposite lateral sides **14A** and **14B**. The inner side **11** includes a bottom layer comprising an absorbent part **16** providing for receiving human discharge including menses, urine and faeces. The absorbent part **16** is flanked by leak guards **18A** and **18B** at each side **14A** and **14B**.

[0040] As is known in the art, the absorbent part **16** is layered and includes a top sheet that comes into contact with the skin of the user, an acquisition distribution layer and an absorbed core all of which are supported by the diaper chassis including the back sheet defining the outer side **13**.

[0041] The front and rear ends **12A** and **12B** define the front and rear parts **20A** and **20B** of the diaper **10**. Each part **20A**, **20B** has respective lateral side wings **22A'**, **22A''** and **22B'**, **22B''** which include adhesive or other fastening elements. In the closed position, wings **22A'** and **22B'** are fastened together to define one lateral part **24'** (at side **14A**) of the diaper and wings **22A''** and **22B''** are fastened together to define the other lateral part **24''** (at side **14B**). In the closed position, the diaper **10** defines leg opening **26A** and **26B** at sides **14A** and **14B**, respectively.

[0042] With particular reference to FIG. **1**, the median section **28** of the diaper **10** interposed between the front and rear parts **20A** and **20B** comprises leg cuffs **30A** and **30B** at each side **14A** and **14B**, respectively which form the openings **26A** and **26B** together with lateral parts **24'** and **24''**.

[0043] Diaper **10** includes a pouch or pocket **32** formed of a top layer or sheet **34** overlying the absorbent part **16**. The layer **34** defines a peripheral edge section **36** thereof that is adhesively mounted to the inner side **11**.

[0044] More particularly, the peripheral edge section **36** is adhesively mounted about and/or on or near the peripheral edge **37** of the absorbent part **16** to be sealed thereto allowing the main portion **38** of the sheet **34** to be loosely resting on the absorbent part thereby forming a pouch or pocket cavity therewith. As such, the edge section **36** can be mounted on an area of the inner side **11** surrounding the absorbent part **16** or can be mounted directly on or near the edge **37** of the absorbent part **17** or both as can be contemplated by the skilled artisan. The purpose of the top layer or sheet **34** is to be adhesively mounted to the inner side **11** of the diaper **10** about its peripheral border edge **36** and to overly the absorbent part **16** and to define a top layer area or main portion **38** which overlies at least a portion of the the absorbent part but is unconnected thereto so as to form a pouch **32** with the inner side **11** including at least a portion of the absorbent part **16**.

[0045] An opening **40** is formed through the sheet **34** about the area where the anus of the user located when worn and leading to the pouch cavity for allowing faecal matter **F** to be collected within the pouch **32** via opening **40** (as represented in FIG. **4**) thereby filling the pouch **32** during user defecation.

[0046] The sheet **34** has a front part **42A** and a rear part **42B** which define the front and rear parts **42A**, **42B** of the pouch **32** (as shown in FIG. **5**) when the sheet **34** is mounted to the diaper **10** as provided herein. The front part **42A** comprises a front recessed portion **44** in the form of a V-like or U-like cut-out defining a pair of laterally elongated fingers **46A** and **46B** at each side **14A** and **14B** respectively exposing a portion **17** of the absorbent part **16** therebetween which is not covered by the sheet **34**.

[0047] As better shown in FIG. **5**, the fingers **46A** and **46B** have inner edges **47A** and **47B** which form part of peripheral edge section **36** and which meet at a common juncture **49** therebetween (see also FIG. **4**). The inner edges **47A**, **47B** and the common juncture **49** form a valley configuration which comprises the portion **17**. The common juncture **49** defines the floor of the valley.

[0048] Portion **17** of absorbent part **16** is so positioned to directly receive urine discharged by the user and as such is direct contact with the user's body corresponding to the genitalia area of the user i.e. the penis/vagina of the user depending on the gender thereof. The foregoing avoids for urine discharge onto open absorption portion **17** to mix with faecal discharge **F** collected in the pouch **32**. Indeed, portion **17** may include additional layers of absorbent material as can be contemplated by the skilled artisan within the scope of the present disclosure.

[0049] As shown in FIGS. **5** and **6**, the rear part **42B** has a generally rectangular configuration and defines a central line **48** flanked by two halves **50A** and **50B**. The opening **40** is formed about the central line area **48** (see also FIG. **1**). The rear edge **52** (see also FIG. **4**) of the rear part **42B** forms part of the peripheral edge section **36**. The fingers **46A** and **46B** define respective front ends **54A** and **54B** and form part of the peripheral edge section **36**. The pouch **36** includes lateral edges **56A** and **56B** (see also Figure **4**) at each lateral side **14A** and **14B** respectively which form part of the peripheral edge section **36**.

[0050] With reference to FIGS. **5** and **6**, diaper **10**, in accordance with an embodiment thereof, includes a tear line **58** along a portion or the full central line area **48**. The central line area **48** runs from juncture **49** to the rear edge **52** and the tear line **58** can run along the full length of the line area **48** or a portion of it. At least one portion of tear line **58** is situated at a portion **60** (see FIGS. **6** and **7**) of the line area **48** that corresponds to the anus of the user, usually the middle part of the line area **48**. As mentioned, the tear line **58** can extend beyond portion **60** defining front and rear portions **60'** and **60''** respectively.

[0051] The tear line **58** defines perforations **62** which allow a user to open a portion of the tear line **58** in order to form an opening **40** having the desired length and the desired position. The user can for example, pull about the tear line **58** on a portion of the sheet **34** at halves **50A** and **50B** or otherwise tear a desired portion of the line **58** in order to form an opening **40** at a desired position along the line area **49**. Since humans have different anatomies, the position of the anus and the configuration of the intergluteal cleft differs from person to person and thus the position and size of the opening **40** may need to vary from one individual to another. Having tear lines allows the caregiver to selectively make an appropriate opening **40** with a desired length and position along the line area **49** depending on the anatomical particularity of the individual wearing the diaper **10**.

[0052] In an embodiment and with reference to both FIGS. **6** and **7**, there are relatively more and larger perforations or longer weak lines **62** at middle portion **60** and relatively lesser and smaller perforations or shorter weak lines **62** at either one of the front and rear portions **60'** and **60''**. Indeed, since in most cases, the opening **40** will most likely fall within section **40**, the greater number of perforations/weak lines **62** provide for tearing the tear line **58** at smaller increments to allow a greater number of possible opening **40** positions than would be required in portions **60'** and **60''**.

[0053] In some cases, the weak lines or perforations **62** may not be sufficiently visible and the sheet **34** may be marked with indications similar to the schematic illustration of FIG. **7** to provide just one example in order to more clearly show where the weak lines or perforations are positioned as well as their numerical concentration.

[0054] Indeed, the tear line **58** may be configured in a variety of ways known in the art within the context of the present disclosure providing for conveniently tearing the sheet **34** to make the opening **40**.

[0055] Sheet **34** defines the outer or top surface **35** of the pouch **32** which will be in contact with the wearer's skin.

[0056] As shown in FIGS. **5** and **6**, the pouch **32** comprises an adhesive **64** on its outer surface **35** at each lateral side of the tear line **58**. The adhesive **64** is provided in the form of two adhesive strips **64A** and **64B** flanking the tear line **58** and running longitudinally along the length of the central line area **48**. In an embodiment, the adhesive **64** is a low-density adhesive configured for adhering to human skin (such as edible glue for example). Each strip **64A**, **64B** adheres to a

respective buttock of the wearer along the inner side thereof within the intergluteal cleft.

[0057] As represented in FIG. 8, the adhesive strips **64A**, **64B** are sealed to the inner sides **S1**, **S2** of the buttocks **B1**, **B2** respectively within the intergluteal cleft **C** bringing the opening **40** towards the anus **A**. In this way, during defecation, feces are blocked from the spilling unto the outer surface **35** by the sealed junction **J1** between adhesive strip **64A** and the inner buttock side **S1** and the sealed junction **J2** between adhesive strip **64B** and the inner buttock side **S2**.

[0058] Turning now to FIGS. 5 and 6 and accordance with an embodiment of the diaper **10**, the outer surface **35** of the pouch **32** also comprises additional adhesive material **66** on the sheet halves **50A** and **50B** for adhering to the outer surface (glute area) the buttocks. In the embodiment, the adhesive material **66** comprises strips of adhesive radially extending relative to the central portion line area **48**. In the example shown here, sheet half **50A** comprises a rear upwardly diagonally extending adhesive strip **66A'**, a middle laterally extending adhesive strip **66A''**, a front downward diagonally extending adhesive strip **66A'''**. The sheet half **50B** comprises strips that are essentially a mirror image of those of sheet half **50B**, namely a rear upwardly diagonally extending adhesive strip **66B'**, a middle laterally extending adhesive strip **66B''**, a front downward diagonally extending adhesive strip **66B'''**. Of course, the skilled artisan will readily appreciate that adhesive **66** can be positioned on surface **35** in a variety of suitable ways within the context of the present disclosure.

[0059] Turning back to FIG. 7, the adhesive **66** provides for adhesively sealing each half **50A**, **50B** to a respective outer surface (glute area) **G1** and **G2** of each buttock **B1**, **B2**, respectively. Thus, the foregoing allows to maintain the sheet **34** in position with the opening **40** near the anus blocking spillage of feces between surface **34** and the buttocks of the user and allowing the feces to flow into the pouch cavity **67** (see FIG. 8) via opening **40**. Maintaining the sheet **34** in place is especially required during movement of the user.

[0060] In an embodiment, the opening **40** (and the remaining untorn tear line **58** if the whole tear line **58** has not been torn) provides for less tension in the central line area **48** (relative to the tension in the sealed edge portion **36**) moving it upwards into the intergluteal cleft **C** further helping position the opening **40** at anus **A** and providing a synergistic effect to the sealing action of the adhesives **64** and **66** as described hereinabove.

[0061] In an embodiment and as shown in FIG. 9, the adhesives **64** and **66** are covered by removal liner films **68** which are removed prior to diaper use.

[0062] In an embodiment and as shown in FIG. 10, the adhesives **64** and **66** are covered by removal sheets **70** which are removed prior to diaper use. In an embodiment each of pair of sheets **70** is positioned on a respective half **50A**, **50B** of the surface **35** without covering the tear line **58** therebetween.

[0063] When the caregiver removes the diaper **10** from the user following defecation and filling of the pouch **32** with faecal matter **F** as shown in Figure, the diaper provides for avoiding any faecal matter from spilling unto surface **35** or anywhere in the inner side **11** thereby not exposing the caregiver to the faecal matter and moreover, the caregiver can fold the diaper with it's the inner sides of its front and rear parts **42A** and **42B** meeting or about the central line area **48** as is convenient to enclose the inner side **11** and close the opening **40**. When folding the adhesives **64** and **66** will provide for adhering to parts of the inner side **11** they interface with thereby sealing the diaper **10** so as to fully enclose the inner side **11** and seal the pouch **32** with only the outer side **13** being exposed. The foregoing provides for exposing the caregivers to the toxic gases of the faecal matter which are a health hazard.

[0064] With reference to FIG. 11, there is shown a top schematic view of a diaper **110** have front and rear ends, **112A** and **12B**, respectively, and opposite lateral sides **114A** and **114B**. As taught in PCT/CA2020/000028, the diaper **110** includes top and bottom layers that are glued together at top and bottom edge portions **116A** and **116B** respectively. As taught in PCT/CA2020/000028 the diaper **110** includes an opening **118** leading to a collection pouch.

[0065] In an embodiment, the strength of the glued portions **116A** and **116B** provide a longitudinal tension **T1** that extends throughout the length **L** of the diaper. This tension **T1** is related to the length of the diaper **110** vs the width **W** of the diaper **110**. In an embodiment, the vertical tension **T1** is two to three times greater than the lateral tension **T2** across the width of the diaper. Having a longitudinal tension **T1** that is greater than the lateral tension is convenient for avoiding the diaper **110** to fold inwardly bringing in sides **114A** and **114B** closed together. Furthermore, the greater vertical tension **T1** provides the C shaped curve to the diaper **110**. As ends **112A** and **112B** are pulled inwardly, the shorter length of the inner layer creates a gap with the longer bottom layer thus forming the collection pouch.

[0066] During manufacturing, the bottom layer will be stretched out more vertically than laterally and then the inner layer will be glued onto the bottom layer providing the desired strength at the glued portions **116A** and **116B** which further provides the tension **T1**.

[0067] Turning to FIG. **12**, when the top layer (or liner) **120** is stretched to be glued onto the bottom layer as explained above, tension waves **122** are created along the longitudinal length of the top layer. The tension waves are formed by selected stretching tension and their position can be selectively adjusted by modulating the stretching tension. It is convenient to cut a portion **124** of the material **120** to form the opening **118** at a tension wave **122** as the material **126** surrounding this portion **124** (and thus the opening **118**) is upwardly rising forming an apex **128** at the opening **118**. This apex finds its way into the intergluteal cleft of the user and thus provides for positioning the opening **118** at the anus.

[0068] Turning back to FIG. **11**, the diaper **110** includes, in an embodiment thereof, central soft elastic elements **130A** and **130B** respectively sewed into the top layer **120** from end **112A** to the opening **118** and from the opening **118** to the end **112B**. These elastic elements **130A** and **130B** form an apex that upwardly extends along their length which like the apex **128** finds its way into the cleft. In an embodiment, the elastics **130A** and **130B** reinforce apex **128**.

[0069] Indeed, the foregoing apex-forming configuration allows for guiding the inner layer **120** into the cleft and thereby positioning the opening at the anus of the wearer. In fact, the inner layer **120** is not fully connected to the bottom layer and thus there is discrepancy in the lateral movement therebetween, this allows a slack to the inner layer **120** for being guided by a reinforced apex into proper position with the apex **128** inserted into the cleft and the surrounding left and right-side material **126** respectively following the curvatures of the left and right side glutes as this is the path of least resistance.

[0070] Once the opening **118** is formed, due to the fact that the material **120** is cut, the longitudinal tension **T1** along the opening **118** disappears, thus the lateral tension **T2** is no longer the same in the material portions on the left and right of the opening **118**, a new lateral tension is formed, represented by **T3**. The tension **T3** is lesser than the tension **T2**. Prior to the formation of the opening, the relationship between tensions **T1** and **T2** was at a 90-degree angle (i.e. length vs. width).

[0071] Due to the tension ratio between **T1** and **T3** and the circular configuration of the opening **118**, there is produced a radial tension **TR** in the left and right material portions **132A** and **132B** surrounding the opening **118**.

[0072] A rectangular area **134** directly surrounding opening **118** is sealed. In an embodiment, the inner layer **120** receives a silicone (a layer, a strip, or embedded etc.) in area **134** which seals this material **120** within area **134** from waves. The silicone in area **34** allows for this area to seal against the skin of the left and right inner glutes along the cleft therebetween. This is convenient, as the material **124** and the opening **118** act as a funnel to receive feces therein during defecation.

[0073] Nevertheless, it is convenient for the area **134** to be completely flat and devoid of any of tension waves which avoids the feces from escaping via tunnels formed between the valleys of the tension waves and the inner gluteal skin on which area **134** seals to. As such, feces are lead directly within the opening **118** due to the apex **128** pushing the opening upwardly close to the anus and the

material **134** sealing the area of the intergluteal cleft surrounding the anus as provided hereinabove. Thus, there is no spillage unto the top face of the inner layer **120**.

[0074] The lateral portions **136A** and **136B** of the diaper **110** are sections where the top and bottom layers are glued together. In order to avoid the above discussed radial waves TR from being formed, wider lateral portions **138A** and **138B** of the top and bottom layer are glued together along a length of the portions **136A** and **136B**. As such, there is less loose material in areas **132A** and **132B** between the opening **118** and the portions **138A** and **138B**. Indeed, the foregoing flattens out the tension waves TR as it pulls back the slack in material **132A** and **132B** between the opening **118** and the portions **138A** and **138B**.

[0075] Moreover, the lateral pulling provided by new T3 when considering the additional tension provided by sections **138A** and **138B** as explained and in conjunction with T1 provides for a convenient opening tension on the opening **18**.

[0076] It should be noted that the tension provided by sections **138A** and **138B** may cause the material portions **132A** and **132B** to curve, since the material is being pulled laterally but also downwardly. In essence, we are anchoring more material via sections **130A** and **130B**. This lateral and downward pull causes the material portions **132A** and **132B** to pull the area **134** away from the skin, and hence, a correction force is required to pull the area **134** upwardly and make sure it seals to the skin. The foregoing “upward pull” is provided by the longitudinal soft elastics **140A** and **140B** which run along the length L of the diaper **110** and are positioned at the lateral edges of area **134**. Additional upward pull is also provided by opposite pairs of soft elastics **142A** and **142B** running along the width of portions **132A** and **132B** respectively between sections **138A**, **138B** and elastics **140A**, **140B**, respectively. Indeed, as shown in FIG. **12** the foregoing provides an angular upward pull tension T4 in areas **126** which correspond to the area **134**.

[0077] The opening **118** can be closed by modifications of the c-shape of the diaper **110** depending on the size of the wearer. In certain cases, the c-shape will be more closed while other times it will be more open. In order to neutralize this variable diagonal low resistance elastics **144** are positioned between the longitudinal ends of the of the opening and the lateral ends **114A** and **114B** thus making sure the opening **118** remains open for receipt of faecal matter at all times as provided herein. As such, elastics **144** compensate for the c-shape cup movement of the diaper **110**.

[0078] Turning to the example of the diaper **110** shown in FIG. **13**, the portions **146A** and **146B** of the inner layer **120** are shown to be cut in a zigzag and thus glued along this zigzag configuration on the bottom layer. Thus, the tension T2 in the area **148** of the diaper **110** will be formed by a series of vectors that are diagonal rather than being in a straight line from end to end, as the zigzag forms oppositely directed diagonal lines **150'** and **150''**, each providing a similarly directed vector across the area **148**. The pitch of this zigzag can be modulated in order to correspondingly modulate the diagonal tension vectors across area **148**. The tension across area **148** will thus be diluted. Indeed, with this zigzag, a much greater but smaller number of waves are created thereby smoothing out the material in area **148** rather than having lesser and larger waves which are more difficult to straighten out. Thus, in this case there are so many waves that they become negligible.

[0079] Turning to FIG. **14**, the contour **152** of the inner layer **120** of the diaper **110** is configured to decrease high tension lines by providing smaller areas of tension. For example, the rear end **112A** can be slightly tapered **154** and the junctions **156** between the end **112A** and lateral sides **114A** and **114B**, respectively can be rounded rather than being a corner, thus softening the tension points. Thus, smaller tension lines provide for smaller waves and these smaller waves can easily be absorbed and become negligible across the top surface **158**. The junctions **156** are cut in a zigzag shape to provide the benefits described for FIG. **13**. As explained for FIG. **11**, the lateral glued end sections **160** at the level of the opening **118** are larger in order to anchor more material for the benefits previously described. In this example, the sections **160** are also in zigzag configurations that are larger (more inwardly glued, thus providing larger triangular indentations) than junctions **56** as well as the front end sections **162** which do not have the opening **118** interposed

therebetween but rather a contiguous uninterrupted area **164**.

[0080] The opening **118** is surrounded by a silicone line **166** that provides an upward vector towards the anus at the same time the tension of sections **160** provide a lateral pull on the opening thereby causing the silicone line **166** to move upwards and outwards towards edges **114A** and **114B** thus sealingly gripping the inner gluteal skin circumscribing the anus thus feces flows into the pocket and not between the inner layer **120** and the skin.

[0081] In the event that a wearer defecates the feces will seal the inner layer or liner **120** with the absorption core making it very likely that if the user subsequently urinates the urine will not be able to reach the core and will spill. The diaper **110** of FIG. **15** solves this issue. An air laid paper **168** is provided at the front end portion of the diaper **110** to quickly absorb the urine and lead it to the absorbent core. The air laid paper **168** is positioned between the liner **120** and the core.

[0082] The diapers can be made of a variety of suitable materials and designed in a variety of suitable configurations within the context of the present disclosure.

[0083] The various features described herein can be combined in a variety of ways within the context of the present disclosure so as to provide still other embodiments. As such, the embodiments are not mutually exclusive. Moreover, the embodiments discussed herein need not include all of the features and elements illustrated and/or described and thus partial combinations of features can also be contemplated. Furthermore, embodiments with less features than those described can also be contemplated. It is to be understood that the present disclosure is not limited in its application to the details of construction and parts illustrated in the accompanying drawings and described hereinabove. The disclosure is capable of other embodiments and of being practiced in various ways. It is also to be understood that the phraseology or terminology used herein is for the purpose of description and not limitation. Hence, although the present disclosure has been provided hereinabove by way of non-restrictive illustrative embodiments thereof, it can be modified, without departing from the scope, spirit and nature thereof and of the appended claims.

Claims

1. A diaper comprising: an outer side and an opposite inner side comprising an absorbent part; and a top layer defining a peripheral border thereof and being sealed to the inner side about the peripheral border thereof and defining a top layer area delimited by the peripheral border overlying at least a portion of the absorbent part and being unconnected thereto, the top layer area, the absorbent part and the peripheral border defining a pouch therebetween, the pouch defining a pouch cavity therein, the top layer area defining a body contacting surface and providing an opening for access to the pouch cavity; wherein when the diaper is worn by a user, the opening is provided to be positioned about the anus of the user for providing for faecal matter to enter the pouch cavity during defecation.
2. A diaper according to claim 1, wherein the opening is provided along a central line area of the top layer area.
3. A diaper according to any one of claim 1 or 2, wherein the top layer area comprises adhesive at each lateral side of the opening, the adhesive providing for adhering the user's skin at an inner side of each buttock within the intergluteal cleft thereby bringing the opening in close proximity to the anus to receive faecal matter directly therein.
4. A diaper according to claim 3, wherein the adhesive is in the form of a pair of longitudinal strips along a longitudinal length of the top layer area so as to be removably sealed along a longitudinal length of the user's skin.
5. A diaper according to any one of claim 3 or 4, wherein the adhesive provides a seal with the user's skin about each lateral side of the anus and the opening substantially blocking faecal matter from flowing onto an outer surface of the top layer area beyond the seal thereby allowing faecal matter to flow into the opening.

- 6.** A diaper according to claim 5, wherein the adhesive for adhering the user's skin at an inner side of each buttock within the intergluteal cleft is covered by a removable liner prior to use.
 - 7.** A diaper according to any of claims 1 to 6, comprising adhesive on the top layer for providing the top layer for adhering to the user's skin directly on each buttock outside the intergluteal cleft so as to maintain the diaper in position during movement of the user.
 - 8.** A diaper according to claim 7, wherein the adhesive for adhering to the user's skin directly on each buttock outside the intergluteal cleft is provided in the form of elongate strips on the top layer for providing substantial areas of the top layer without any adhesive.
 - 9.** A diaper according to any one of claim 7 or 8, wherein the adhesive for adhering to the user's skin directly on each buttock outside the intergluteal cleft is covered by a removable liner prior to use.
 - 10.** A diaper according to any one of claims 1, wherein the top layer area comprises a tear line for being selectively torn along a length thereof to selectively form the opening.
 - 11.** A diaper according to claim 10, wherein the tear line is formed along a central line area of the top layer area for being selectively torn along a desired position of the central line area and along a desired position of the central line area.
 - 12.** A diaper according to any one of claim 10 or 11, wherein the tear line comprises weak lines or perforations, the tear line defining a central portion interposed by front and rear portions, wherein the central portion has a greater number of the weak lines or the perforations than either one of the front and rear portions.
 - 13.** A diaper according to any one of claims 1 to 12, wherein the top layer provides for allowing a front part of the absorbent part to be uncovered and thus in direct contact with the user for receiving urine discharge to be absorbed, thereby substantially avoiding urine discharge from into the pouch and mixing with the feces collected therein.
 - 14.** A diaper according to claim 13, wherein the pouch defines a front part and a rear part, the front part defining a pair of front separate fingers defining an inwardly recessed portion therebetween exposing the front portion of the absorbent part in direct contact with the user for receiving urine discharge to be absorbed.
 - 15.** A diaper according to claim 1, wherein the top layer comprises adhesive for adhering to the users skin and for adhesively sealing the inner side when folding the diaper after use thereby sealing the pouch containing the collected faecal matter.
 - 16.** A diaper according to any one of claims 1 to 15, wherein the inner side defines a length and width thereof, wherein the inner side comprises a longitudinal tension along the length thereof that is greater than the a lateral tension along the width thereof.
 - 17.** A diaper according to any one of claims 1 to 16, wherein the top layer area defines a length thereof and forms waves defining longitudinal peaks and valleys along the length thereof, wherein the opening is provided at a selected one of the longitudinal peaks.
 - 18.** A diaper according to any one of claims 1 to 17, wherein the peripheral boundary of the top layer defines lateral sides thereof, wherein at least a portion of the lateral sides of the peripheral boundary are cut in a zigzag configuration.
 - 19.** A diaper according to any one of claims 1 to 18, wherein the top layer comprises soft elastic elements running along a length thereof.
 - 20.** A diaper according to any one of claims 1 to 19, wherein the top layer area comprises silicone circumscribing the opening.
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